

LexSetu

Nyaya Setu — The Citation-Verification Pivot

A free, open-source legal AI assistant for India, being rebuilt around one capability: proving that a citation in a legal filing actually says what it's claimed to say — before a court has to find out the hard way.

Document date	22 September 2026 (revised)
Sprint window	22-24 September 2026
Hackathon dates	25-26 September 2026
Prepared for	Team LexSetu, Buildablelabs

What changed in this revision: every milestone below is written to be read on its own — each item states not just what gets built, but why it matters and what it looks like once it's working. Four items missing from the previous version (step-by-step draft walkthrough, contradiction diff view, clickable citation trail, and a working local-only mode toggle) are now explicitly placed inside their milestones. Page 9 is a full feature-by- feature coverage map, so nothing discussed across this project is left unaccounted for.

What we're building

LexSetu is a free, open-source legal AI assistant for India that is repositioning from a drafting tool into a **citation-verification layer**: it extracts every case citation in a filing or draft, checks whether the cited case exists and whether the quoted paragraph or holding actually matches the judgment's text, and returns one of three honest states — **Verified**, **Mismatch**, or **Not found in indexed corpus** — instead of a silent guess.

The trigger for this pivot is concrete, not speculative: on 2 July 2026 the Supreme Court in *Pooja Ramesh Singh v. J&K Bank* (2026 INSC 668) set aside NCLT/NCLAT orders after finding that of six judgments cited in an AI-assisted filing, three did not exist and the other three were misquoted. The Court directed the Bar Council of India to frame AI-use norms and treated the unverified citations as advocate misconduct. Its own November 2025 white paper already asks judges to verify AI output — but gives lawyers no equivalent tool for their own filings, in a system where district judges average roughly 14-17 minutes per case. LexSetu's bet is that this specific gap — not faster drafting — is what lawyers, students, judges, and legal-aid workers will actually value.

Current state — what exists today vs. what's being added

CAPABILITY	STATUS	NOTES
Draft generation over 1,841 template documents (Qdrant + FastEmbed)	Built	Core RAG pipeline, chunked at fixed 250 words / 40 overlap
Two-pass review engine (doc-type check, placeholder/heading extraction, re-retrieval comparison)	Built	Real second-pass verification, but only against templates
Live case-law search (Indian Kanoon → scrape → CommonLII fallback)	Built, disconnected	Exists, but not wired into drafting or grounding — this is what the sprint fixes
Contradiction scanner between two documents (JSON output)	Built	LLM-based comparison, functional but shown as raw JSON today
Groq multi-model cascade + Ollama offline fallback	Built	6-model cascade with local fallback — the raw material for a "local-only" mode
Judgment-grounded retrieval (separate corpus, structure-aware chunking)	Building this sprint	New — open-india-law + IndiaCode integration
Citation classifier + entailment/paragraph-match verification	Building this sprint	New — the core of the pivot
Verify-a-Filing mode + Citation Integrity Report + exportable table	Building this sprint	New flagship feature
IPC→BNS / CrPC→BNSS / Evidence→BSA mapping table	Building this sprint	Deterministic, curated — not LLM-generated
Audit trail, precedent-validity flag, self-building corpus	Roadmap	Explicitly out of sprint scope
Production infra, institutional integration, paid tier	Roadmap	Post-hackathon, 1-4 months out

Sprint milestones — 22-24 September

Three build days remain before the hackathon. Each milestone is written to stand alone: read just one, and you should understand exactly what it builds, why it exists, and how you'd know it worked — without needing the rest of this document.

S1 Engine Foundation

Mon, 22 Sept

In one line: give LexSetu a second memory — of real judgments, not just templates — so it has something real to check a citation against.

Today LexSetu can only retrieve draft templates; it has no indexed body of actual case law. Every later milestone — verification, reports, confidence colors — depends on this existing first. Nothing in S2, S3, or S4 can start meaningfully until this is done.

WHAT SHIPS

- **Judgment corpus integration.** A curated slice of the open-india-law dataset (CC BY 4.0) plus IndiaCode, loaded into a brand-new Qdrant collection named `judgments` — kept completely separate from the existing 1,841-template collection, so drafting and verification never interfere with each other or accidentally cross-contaminate results.
- **Structure-aware, parent-child chunking.** Templates are chopped into fixed 250-word blocks, which is fine for short forms but wrong for a 40-page judgment. Judgments instead get split along their own real structure — numbered paragraphs, section boundaries — into small, precise child chunks for matching, while each child stays linked to its full parent (the whole judgment or section). This means retrieval can hand back complete context around a match, not an isolated sentence ripped out of the middle of a paragraph.
- **`search_judgments()`.** A new, separate retrieval function (distinct from the existing template search) that takes a query — a citation, a legal question — and returns the most relevant judgment passages, each tagged with its court, date, and paragraph number.

What this looks like to the user: nothing yet — this milestone is entirely invisible plumbing. No UI changes today. That's expected and fine; S2 is where it becomes visible.

Exit gate: write 10 manual test queries against real content in the indexed slice (things you can verify by hand — a known case name, a known legal issue). Run them through `search_judgments()`. At least 7 of 10, ideally all 10, must return the correct judgment in their top-5 results before moving on.

S2 — Verification Core + Verify-a-Filing

S2 Verification Core + Verify-a-Filing

Tue, 23 Sept

In one line: the actual product — upload any filing and get back an honest answer on whether its citations are real and accurate.

This is the flagship feature and the entire reason for the pivot: the Supreme Court just penalised a lawyer for exactly the failure this milestone catches. S1 built the memory; S2 is what does something useful with it. This is the hard gate of the whole sprint — if this milestone slips, the sprint has no product to show, regardless of what else ships.

WHAT SHIPS

- **Citation classifier.** Scans any piece of text and detects exactly where a case citation appears, so the system knows precisely which sentences need checking rather than treating the whole document as one undifferentiated block.
- **Entailment / paragraph-match check.** For every citation the classifier finds, an LLM-as-verifier step compares what the filing claims that citation says against the actual judgment paragraph retrieved by S1's index, and judges whether the claim genuinely follows from the real text — not whether it merely sounds plausible.
- **Three-state verification model.** Every checked citation lands in exactly one of three states, never a binary: **Verified** (the case exists and the paragraph/holding matches), **Mismatch** (the case exists, but what's quoted doesn't match), or **Not found in indexed corpus** (no match in the current index — deliberately never shown or worded as "fake," since the corpus is a curated slice, not the whole of Indian case law).
- **Inline confidence indicator (generalised).** The same green / amber / grey visual language extends beyond citations to an entire draft: green marks content grounded in a verified source, amber marks content inferred from a template without independent verification, and grey marks a [Not Provided] placeholder still waiting on a real fact. A reader can tell at a glance which parts of any output to trust outright and which to double-check themselves.
- **Clickable citation trail.** Every citation in a checked or generated document becomes a clickable chip. Clicking it opens the actual retrieved source paragraph side-by-side with the claim, so the user verifies the match themselves in one click instead of taking the system's word for it.
- **Verify a Filing.** The headline feature: upload any petition or written submission — not necessarily one LexSetu drafted — and the system extracts every citation in it, runs each one through the check above, and returns a complete Citation Integrity Report.
- **Exportable Citation Verification Table.** The report exported as a clean table — case, court, date, paragraph, source link, status — ready to print or attach directly to a filing as supporting evidence of due diligence.
- **Coverage banner.** A permanently visible line stating exactly what's indexed and as of when ("Indexed: X judgments, courts Y, updated Z"), so the system never silently implies broader coverage than it actually has.

What this looks like to the user: they upload a document, wait a few seconds, and see a report where every citation carries a coloured status, opens its source text in one click, and can be exported as a table for their own records or for the court.

Exit gate (hard — the gate of the sprint): Verify-a-Filing must work standalone, end-to-end, on at least 2 real uploaded filings. If this is not true by the end of the day, S3 the next morning becomes a bug-fix day only — nothing in S3 ships ahead of this being solid.

S3 — Persona Features + Validation

S3 Persona Features + Validation

Wed, 24 Sept

In one line: turn the verification engine into something each type of user gets direct, daily value from — and prove the numbers behind it instead of promising them.

S1 and S2 build the engine; S3 is where the same engine becomes four different front doors — one each for the lawyer, the student, the judge, and the paralegal — and where the credibility benchmark that goes into the pitch actually gets run, not just claimed.

WHAT SHIPS

- **Placeholder → questionnaire.** An unfinished draft today just shows raw blanks like [DATE] or [PARTY NAME]. This turns those into a short structured questionnaire ("FIR number? Date of arrest? Prior bail sought?") so completing a draft feels like answering a form, not hunting through brackets.
- **Citation-format resolver.** The same case can be cited as AIR, SCC, INSC (neutral citation), or by its LiveLaw name. Paste any of these in, and the resolver returns the one canonical case record — aimed at law students, for whom moot-memorial citation-format mismatches are a common, entirely avoidable error.
- **IPC→BNS / CrPC→BNSS / Evidence→BSA table.** A curated, deterministic lookup of the ~50 most-used old-law sections mapped to their new-law equivalents. Deliberately not LLM-generated for this table — a fixed lookup has zero hallucination risk, where a model doing the same mapping does not.
- **Judge-facing bench note.** A one-page auto-generated summary for a judge: the issues in the matter, a list of dates, and both sides' cited precedents with their verify status already attached — built so a 14–17-minute hearing slot doesn't have to include manual citation-checking on top of everything else.
- **Local-only / offline mode, made visible.** The Ollama fallback already exists as an engineering resilience feature; this gives it a name and a UI toggle — "local-only mode: client data never leaves this device" — turning a backend fallback into a trust feature a lawyer can point to directly when client confidentiality is the concern.
- **Step-by-step draft walkthrough.** Instead of dumping a finished draft as one wall of text, this breaks it into a guided sequence — one section at a time, explained in plain language — so a student or a self-represented litigant understands what each part of the document is doing, not only what it says.
- **Side-by-side contradiction diff view.** The existing contradiction-scanning backend already finds conflicts between two documents but returns raw JSON. This gives that output an actual visual diff — the two conflicting passages shown next to each other, highlighted — instead of a text dump only a developer could read.
- **Regional-language (Hindi) grounded mode.** A Hindi translation of an already-verified answer, with the original source attribution carried over intact, so translation itself never becomes a second place for errors to sneak in.
- **Video explainer** (*stretch — only if every item above is done with time to spare*). A grounded text answer is turned into a script and sent to a video-generation API (HeyGen/D-ID) to produce a short explainer clip — positioned deliberately as the closing demo beat, not the headline, since trust has to be established before delight for this audience.
- **20-citation planted-error benchmark.** Built and run today: 5 fake case names, 5 real cases cited with the wrong paragraph, 5 real cases with a misstated holding, and 5 correct citations — all run through the S2 engine to produce two real, written-down numbers: detection rate and false-positive rate.
- **Demo golden-path freeze.** The exact filing and queries used in the live demo are pre-cached today, so the presentation never depends on a live model call succeeding in real time in front of the judges.

What this looks like to the user: different things to different people — a lawyer sees a questionnaire and a local-only toggle, a student sees a citation resolver and a guided walkthrough, a judge sees a one-page bench note. Same engine underneath, four different front doors on top.

Exit gate: the benchmark's two numbers exist and are written down; the golden-path demo has been rehearsed once, end-to-end, with zero errors.

Fixed from the previous version: the step-by-step walkthrough, the contradiction diff view, and the local-only mode toggle were discussed earlier in this project but had dropped out of the milestone list. All three are now explicitly built here, in S3.

S4 — Rehearsal & Presentation

S4

Rehearsal & Presentation

Fri-Sat, 25-26 Sept

In one line: make sure nothing that worked in a quiet room breaks in front of the judges.

A real feature that fails live because of venue wifi or a rate limit reads worse than never having built it at all. This milestone exists purely to remove that risk — no new features are written from this point on.

WHAT SHIPS

- **"Catch the hallucination" live demo beat.** The fixed opening of the three-minute presentation: a raw LLM is asked the same question and hallucinates a citation, shown side-by-side with LexSetu's grounded, checked answer catching the exact same error. This is the single clearest way to make the pitch's stakes obvious in the first 30 seconds, before any feature explanation begins.
- **Environment check.** Qdrant, Groq keys, and the offline fallback all confirmed working on the actual venue network — not just at home or in a hotel room the night before.
- **Two full rehearsals** of the fixed demo script before presenting, run exactly as they'll be performed on stage.
- **Q&A preparation.** The hardest questions judges are likely to ask (accuracy, liability, data licensing, "why not just ChatGPT") assigned to specific team members in advance, each with a prepared, honest, two-line answer.

What this looks like to the user: the actual presentation — nothing new to explain, only what was already built and rehearsed.

Exit gate: two clean rehearsals with zero live-call failures; the offline (local-only) path tested and working at least once.

Cut-line — what gives first if a milestone runs behind

If a gate above is missed, drop items in this exact order — and only this order — starting from the least essential: video explainer → Hindi mode → judge bench-note view → step-by-step draft walkthrough (fall back to today's single-block draft) → contradiction diff view (fall back to the existing raw JSON output) → local-only mode UI (keep the underlying Ollama fallback running, drop only its toggle and messaging) → citation-format resolver → IPC↔BNS table (ship 20 rows instead of 50, clearly labeled partial) → placeholder questionnaire (show the raw placeholder list instead) → clickable citation trail (fall back to a plain, non-clickable table) → the confidence- indicator's generalisation beyond citations (fall back to citation-only colouring) → live filing upload in Verify-a-Filing (fall back to a fixed set of pre-loaded demo filings). The S1 retrieval engine and the S2 three-state verification core are never cut under any circumstance — they are the product itself, not a feature on top of it.

SECTION 3

Success criteria

Five measurable targets the sprint is accountable to — not aspirations, numbers that get checked against reality before the demo.

#	Criterion	Target
1	Detection rate on 15 planted-error citations	$\geq 80\%$ (12 of 15 or better)
2	False positives on 5 correct citations	0 of 5 — any hit is root-caused before the demo
3	Citation Integrity Report latency (5-citation filing, end-to-end)	≤ 20 seconds
4	Coverage banner accuracy	Exact match to the live indexed collection count, no hardcoded numbers
5	Red-line copy audit	Zero outcome-prediction language anywhere in the UI or demo script

Design rules that hold across every milestone

Two rules apply everywhere in this document, not to one milestone alone. First: a result is always one of three honest states (Verified / Mismatch / Not found in indexed corpus) — a small demo corpus means "not found" must never be shown or narrated as "fake." Second: every feature is tested against one question before it ships — does it show a *verifiable fact*, or does it suggest a *decision*? Outcome prediction, likely-verdict scoring, or anything resembling it is a hard red line, not a missing feature — a deliberate exclusion that stays deliberate even under pressure to impress a panel.

Post-hackathon roadmap

What the hackathon build deliberately does not attempt — kept visible as "what's next," never folded into the sprint under time pressure.

R1		2-4 weeks
Harness, audit trail, Hindi mode, precedent-validity flag		
Entry	Hackathon demo delivered; benchmark detection rate ≥ 80% confirmed	
Deliverables	Formal tool-calling harness with a completion gate (case lookup, statute lookup, citation verifier, draft generator as distinct callable tools); full audit trail; full Hindi mode; precedent-validity flag	
Exit metric	Audit trail covers 100% of Verify-a-Filing calls; Hindi mode passes native-speaker review; validity flag tested on 20 known-overruled cases	
Depends on	Sprint's S1/S2 judgment-grounding pipeline	
R2		1-2 months
Corpus growth, production infra, security		
Entry	R1 shipped and stable	
Deliverables	Opt-in, reviewed user-contributed corpus ingestion; scheduled corpus refresh; production infra with rate-limit-aware queueing across the Groq cascade; OWASP-grade security pass	
Exit metric	Corpus refresh runs unattended weekly for 4 consecutive weeks; security findings closed	
Depends on	R1's audit trail	
R3		2-4 months
Institutional pilot, licensing, paid tier		
Entry	R2 shipped; validation calls (below) completed positively	
Deliverables	Tele-Law / State Legal Services Authority pilot; law-school licensing package; paid firm tier (audit export, higher limits, team seats)	
Exit metric	One signed pilot MOU or LOI; one paying design-partner firm	
Depends on	R1's audit trail — institutional pilots will ask for it	

Outstanding validation, in parallel with the sprint

The judge/registry reasoning behind this plan is source-backed (Supreme Court white paper, pendency statistics, Kerala HC policy). The lawyer, student, and paralegal feature mapping is reasoned on top of the same sources, not independently validated. Three short calls — a practising advocate, a law student, a DLSA/paralegal volunteer — each asked one question, *"How do you verify citations before filing, and how long does it take?"*, should happen before the hackathon and can reprioritise items within S3 and the roadmap above.

Full feature → milestone coverage map

Every feature discussed for this project, and exactly where it lives. Nothing on this list is unaccounted for.

Feature	Milestone	What it needs
search_judgments() + separate Qdrant collection	S1	open-india-law slice + IndiaCode ingested
Structure-aware, parent-child chunking	S1	Chunking logic for judgment text
Citation classifier + entailment check	S2	Depends on S1 retrieval
Three-state verification model	S2	Depends on entailment check
Inline confidence indicator (green/amber/grey, whole-draft)	S2	Generalises the three-state model beyond citations
Clickable citation trail	S2	UI layer on top of the verification core
Verify-a-Filing mode	S2	Upload endpoint + citation extraction
Exportable Citation Verification Table	S2	Report assembly
Coverage banner	S2	Live collection count
Placeholder → questionnaire	S3	Existing placeholder detection
Citation-format resolver (AIR/SCC/INSC/LiveLaw)	S3	S1 judgment corpus
IPC→BNS / CrPC→BNSS / Evidence→BSA table	S3	Manual curation, no dependency
Judge-facing bench note	S3	Depends on S2 report assembly
Local-only / offline mode (named, with toggle)	S3	Existing Ollama fallback
Step-by-step draft walkthrough	S3	Existing draft generation
Side-by-side contradiction diff view	S3	Existing contradiction scanner
Regional-language (Hindi) grounded mode	S3	Stretch — depends on S2 being solid
Video explainer	S3	Stretch, closing beat only — external video API
20-citation planted-error benchmark	S3	Depends on S2 engine
Demo golden-path freeze / pre-warmed cache	S3	Depends on the finished demo filing
"Catch the hallucination" live demo beat	S4	Depends on S2 being demo-ready
Self-building corpus (user-contributed, tagged, versioned)	Roadmap R2	Out of sprint scope
Audit trail (retrieval → generation → verification log)	Roadmap R1	Out of sprint scope
Precedent-validity flag (still good law or not)	Roadmap R1	Out of sprint scope
Production scalability (managed vector DB, rate limiting)	Roadmap R2	Out of sprint scope
Formal 5-tool harness architecture	Roadmap R1	Out of sprint scope
Tele-Law/SLSA integration, law-school licensing, paid tier	Roadmap R3	Out of sprint scope

LexSetu isn't trying to be an "AI lawyer." It's trying to be the one thing a court just said, on the record, that the profession doesn't yet have: a way to check a citation before it becomes a problem in front of a judge. Every milestone above exists to make that one sentence true, in a working product, within three days.